

Complete questions #1 - #6, #8, #13 - #15

1. What is a GUI?

A graphical user interface. It uses components such as frames, buttons, and labels to communicate with the users.

2. Explain how code is executed in an event-driven application.

A user does an interaction, such as a button click, and the GUI responds by executing a method called an event handler. It determines what was clicked and executes the code related to the button clicked.

3. Can components be added directly to a frame? Explain.

You need to add components onto the content pane / JPanel, which is placed directly into the frame.

4. Can a label respond to events?

No? The text cannot be changed by the user, however, if paired with a button, the label can change what is displayed when button is pressed.

5. Why do you think GUI needs to be run from an event-dispatching thread?

Because the thread makes sure that one event-handler finishes executing before it starts the next one.

6. What is the difference between a label and a button

A label cannot be interacted with by a user, while a button can be clicked to communicate with the application. A button can respond to interaction from a user.

8. List three ways to control the layout of a content pane

Adding borders, using a layout manager (FlowLayout, BorderLayout, or GridLayout) , or setting alignments.

13. What must first be done with numeric data typed in a text field before it can be used in a calculation?

Information in a text field is a string, it must be converted in a double through `parseDouble` to be used in a calculation, and then turned back to a string to be displayed in a JLabel.

14. What is the value of num1 in the last statement below?

15. An application prompts a user to select a name from a list of six names. Which is a better component choice: a text field or combo box?

A combo box, because it condenses the list, making it easier for the user to select their answer and making the application more organized.